FUNDAMENTALS OF AI & ML

Smart study planner using Python

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INTRODUCTION

In today's academic environment, students often struggle to manage multiple subjects, assignments, and deadlines effectively. Poor time management can lead to stress, missed deadlines, and inefficient studying.

This project presents a **Smart Study Planner**, a Python-based system that helps students organize their study schedule intelligently using rule-based logic inspired by Artificial Intelligence concepts.

Problem Statement

Students face difficulty in deciding:

- Which subject to study first
- How much time to allocate to each subject
- How to manage subjects with different difficulty levels and deadlines

This lack of structured planning leads to inefficient preparation and increased academic pressure.

OBJECTIVE

The objective of this project is to:

- Develop a simple system that prioritizes subjects
- Help students create an effective study plan
- Simulate intelligent decision-making without using Machine Learning
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Proposed Solution

The solution is a **rule-based Smart Study Planner** that:

1. Takes input from the user:
    - Subject name
    - Number of days left until exam
    - Difficulty level
2. Applies a **priority scoring formula**:
    - Subjects with closer deadlines get higher priority
    - Subjects with higher difficulty require more attention
3. Sorts subjects based on priority
4. Generates a study plan with suggested study time

Key Decisions Made

- Used **rule-based logic instead of Machine Learning** due to beginner-level constraints
- Chose **priority scoring** as the core mechanism for decision-making
- Kept the system simple using only Python basics for faster development

Challenges Faced

- Designing an effective priority formula
- Balancing simplicity and usefulness
- Ensuring the output is practical for real-life use

Learning Outcomes

Through this project, I learned:

- How to break down a real-world problem into logical steps
- Basics of **AI-inspired systems (rule-based decision making)**
- Importance of clean and structured code
- How to design user-oriented solutions